



Framework proposal to support the suppliers' selection of Humanitarian assistance items: a Flood Case Study in Brazil

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Abstract

Humanitarian Logistics (HL) is comprised of processes and involved systems in the mobilization of people, resources, and knowledge to help affected communities when they are faced with natural disasters. In this study, the Reference Task Model (RTM) provides an overview of these processes and supports Business Process Management (BPM). This article aims to evaluate from a proposed framework the selection of suppliers to guarantee indispensable material resources in the fastest way. We apply a BPM procedure to support the supplier selection process following a flood disaster. We describe each of the stages that make up the proposal of the BPM life cycle applied to the HL. We employ some tools as Balanced Scorecard (BSC) to achieve consensus on objectives, indicators, targets, and actions to be defined for a disaster situation. For balancing the allocation of supplies, the network flow problem is adapted for the quantitative model. That model contains the variables of time, demand, and capacity, and includes the set of adequate suppliers. For the application, we describe a flood disaster case study from a state located in southern Brazil. The main results of the application of the proposed framework are obtained from an optimized holistic view; they represent the selection of suppliers of humanitarian items and consider delivery times, resources, and deprivation costs. One contribution of the proposed framework is the ease of its implementation from process-based technologies and its emphasis on being strategy focused. Further, it concentrates experiences and good practices from humanitarian organizations.

Keywords Humanitarian Logistics · Business process management · Network flow · Balanced scorecard

1 Introduction

In disasters, material consequences can be exemplified by the destruction of health facilities, educational institutions, housing units, and public and private infrastructure, among others.

Similarly, human damages are numerous that can be highlighted in the cases of deaths, wounded, sickness, homeless, displaced, and missing people.

The operational centers of Médecins Sans Frontières (MSF), located in Amsterdam (MSF's Operational Centre Amsterdam, or MSF-OCA), coordinate projects in about 25 countries worldwide; they provide medical assistance to victims of conflict, natural disasters, epidemics, and to those excluded from health care. According to van der Laan et al., (2016), MSF is one of the first humanitarian organizations to realize that logistics is at the core of its operations. Further MSF recognizes that logistics planning can be difficult and time consuming due to the lack of standardized data, the lack of proper logistics support systems, and the lack of metrics for performance measurement and improvement. As stated by MSF, the supply chain in the field is time-consuming and error prone. Although many improvements have been made, the organization's objectives are not delivering as they should. MSF cites additional weaknesses: a lack of foresight, not being focused on the timely delivery of goods, the lack of control for medical local purchases, the lack of clarity between med and logs, and using tools that are not sufficiently modern; all of these weaknesses influence time of service (van der Laan et al., 2016). Moreover, the organization's logistics strategy states their key objective is to have an improved ability to consistently deliver the right goods when needed; this requires forecasting, sourcing, data analysis, reception, and controlled dispensing (van der Laan et al., 2016).

This context suggests the introduction of logistics planning for the development of tasks to assess disaster situations, the purchase of stock relief items, and their storage and respective transport. Humanitarian Logistics (HL) studies problems related to natural disasters, manmade disasters, complex emergencies, conflict situations, and war (Oloruntoba & Gray, 2006).

Humanitarian supply chain activities, including assessment, procurement, warehousing, and transport, organize the flow of people and materials appropriately and promptly in the relief chain, with the main objective of serving the greatest number of people (Beamon, 2004; Ghorbani & Ramezani, 2020; Zejli et al., 2012).

HL is constituted by processes and systems involved in mobilizing people, resources, and knowledge to help vulnerable communities affected by natural disasters or complex emergencies. It aims to serve as many people as possible, avoid lack and waste, organize several donations that are received in these cases, and especially act within a limited budget (Thomas, 2004).

In attendance to the response in disasters, many variables are considered; the time factor, in general, is a main parameters to be taken into consideration. Decision-makers must be able to react quickly and appropriately to the disaster events.

Decision-making required in emergency events presents several challenges that need to be handled according to its specificity, which might be represented according to research methods, analysis procedures, information system, as well as a performance of routine processes and case studies (Salam & Khan, 2020). The control of a disaster situation may be supported by process management which understands all work done to deliver products/services to beneficiaries. This approach is called Business Process Management (BPM) and its application starts at a level strategic to the level that performs the work. Then, it is subdivided into sub-processes that must be performed by one or more activities (workflows) within the functions specified by HL. The activities, in turn, can be decomposed into tasks and, forward, in scenarios of performing tasks and their respective steps (ABPMP, 2013).

To accelerate the modelling/design of the process, it may be helpful to adopt reference models for the definition of the organization's main processes. In the HL case, the Reference Task Model (RTM), proposed by Blecken (2010), is used to incorporate experiences of emergencies that have occurred in the world.

According to Becker et al., (2000), Blecken (2010), and Widera & Hellingrath (2011), using process models offers a set of benefits to organizations since they improve transparency and support the harmonization of relations between the various sectors. It also serves to strengthen the institutional capacity and to manage its growth. Few recent works related to the management of emergency processes have been presented to the humanitarian operations and the management of disaster response (Blecken, 2010; Widera & Hellingrath, 2011).

Balcik & Ak (2014) propose a framework for selecting suppliers for emergencies based on a framework of agreements aimed at ensuring fast and economical procurement. Ghorbani & Ramezani (2020) recommend the integration of transporter and supplier selection to prevent shortages in the supply of humanitarian items. Also, Kim et al., (2019) developed a hybrid multicriteria decision-making model for selecting logistics service providers in the commercial sector. Thus, Hu & Dong (2019) recommend the use of pre-positioning strategies that set discounts on the prices offered by suppliers based on the order quantity and lead time required as well as the physical inventory.

The framework proposed from the RTM model supports the coordination of most of the operations developed by the main humanitarian organization in the state of Santa Catarina, Brazil, called *Defesa Civil do Estado de Santa Catarina* (DCESC), and these are focused on strategic plans that are periodically evaluated. In this paper, the framework will be applied to support the selection of suppliers of humanitarian assistance items. Thus, about the researched proposals for the selection of suppliers in emergencies, some application differences can be observed. Due to Brazil's continental dimensions, the geographical position of the most vulnerable regions, and the large number of suppliers, it is difficult to apply the criterion of procurement agreements, integration of the selection of the transporter and suppliers, and the use of hybrid models of multicriteria decision-making. Here, to select suppliers a pre-positioning strategy was used that considered an optimized deprivation cost related to delivery time. Other publications analyze more sophisticated pre-positioning strategies that consider risks, the stochastic nature of the problem, and the dynamics of emergencies that could be implemented proportionately with a focus on a holistic and integrated view of the processes that support humanitarian operations.

Santa Catarina is one of the states of Brazil, covered by several hydrographic basins. Many of its municipalities were founded on the banks of the rivers, leading to scenarios of sudden and gradual floods, landslides, droughts, tornadoes, and fogs. In July 1983, five days of heavy rains made the rivers rise more than 15 m and this event was considered one of the greatest natural tragedies in the state. In all, 51 deaths occurred and approximately 198,000 persons were left homeless or displaced. The situation made it necessary to supply food, medicines, water, clothes, and even gasoline. Main access routes were flooded, and some municipalities were isolated (Zimmermann, 2011).

Considering the great blow from this event for the state and the location of the suppliers spread over the immense territory of Brazil, this article aims to evaluate the selection of suppliers to guarantee that indispensable material resources can be provided promptly. The

proposed framework recreates a scenario that mirrors this event, so current reality could accommodate should a similar event occur.

The proposed framework is supported by a BPM procedure, complemented by the strategic tool Balanced Scorecard (BSC), focused on the RTM reference model, and defined by an optimized transport flow. The main contributions of this paper are twofold. First, we present the “BPM lifecycle” focused on the RTM model applied to the HL processes. Secondly, through a case study, we empirically explore each of the stages that make up the proposal of the BPM life cycle applied to the HL, using some tools as Balanced Scorecard (BSC) to achieve consensus on objectives, indicators, targets, and actions to be defined for a disaster situation, and, using the “flow problem” on networks adapted to HL to choose suppliers who deliver the supplies as soon as possible.

This paper is structured in more than 4 topics, besides this first one. The next topic presents the theoretical background, focusing on the Reference Task Model and the Framework applied to HL. The third topic presents the case study’s description; we deal with the use of BPM in the case study and present the results. In the next topic, we discuss the main results. Finally, the last topic presents the conclusions and recommendations.

2 Theoretical background

2.1 Reference Task Model

Few and important contributions related to HL in modelling processes can be found in the literature. For example, Tufinkgi (2006) describes a “Logistics Process Model in case of International Catastrophe”, McGuire (2006) develops a framework for the management of the supply chain of health products supplies provided as humanitarian assistance in complex emergencies, and Blecken (2010) suggests the Reference Task Model (RTM).

According to Blecken (2010), in the humanitarian research context, the supply chain processes are not properly documented, and information systems, which could be used to accelerate, automate, and measure these processes, have neither been described nor evaluated. In this context, we proposed the RTM applicable to emergency general tasks, in the supply chain management of humanitarian organizations.

In the construction of this model, Blecken (2010) developed Fig. 1 to guide RTM users to the navigation model and to identify its various layers.

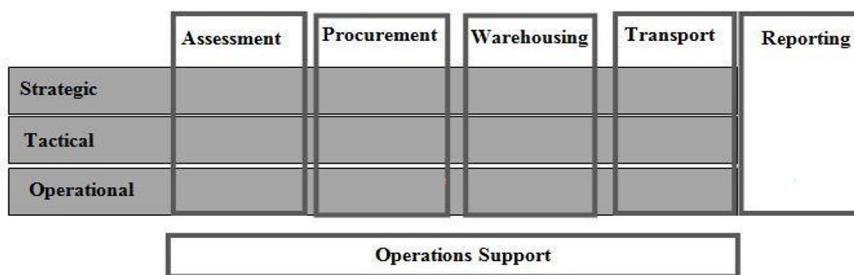


Fig. 1 Framework of the Reference Model for Humanitarian LogisticsSource: Blecken (2010)

The RTM supports the standardization of tasks performed by the organization and its humanitarian partners. The model distinguishes about 120 tasks in the humanitarian supply chain along two axes: the hierarchical decomposition of the planning horizon of strategic tasks, tactical and operational, and a division in the functional axes related to the assessment, procurement, warehousing, transport, reporting, and support operations.

Decisions on the supply chain structure are made at the strategic level, where the time horizon can go beyond two years. At the tactical level, which covers a time horizon from six months to two years, the entire supply chain is planned and optimized. Optimized usage of resources, which are allocated at the tactical level, is the main objective at the operational level. Each task at this level extends up to six months.

Each task is described as well as its alternatives of parameters; those alternatives are indicated when applicable. Explanations are made regarding possible successors and predecessors. Each task is specified as a “generic” or “specific” type; in other words, a generic task is not oriented to a concrete disaster, and a specific one is triggered by a disaster and/or is driven directly to an affected community (response phase and immediate recovery).

The reference model application includes the tasks specialization in case of specific usage, including assigning responsibility and accountability as well as defining information flows. For this, Blecken (2010) used an array of responsibilities, which connects the activities or tasks to people or resources and checks the assignment of duties and responsibilities of the tasks to the humanitarian supply chain agents.

The RTM was developed with the participation of more than 30 humanitarian organizations that reflect the activities of the real world in the management of the humanitarian supplies chain.

2.2 Framework: Procedures Applied in the HL

Frameworks are structures that integrate different projects and/or systems and can represent the various activities and/or purposes of HL (Stewart & Ivanov, 2019). In particular, they are employed as follows: communicating ideas and findings to a broader community, among academics or between academia and industry; making comparisons between different situations and approaches; defining the domain or the limits of a situation; describing the context or arguing the validity of discovery; and supporting the development of procedures, techniques, or methods and tools (Shehabuddeen et al., 2000). There are several proposals to support the main HL processes, such as the framework for comparing Operational Research models applied to HL (Gösling & Geldermann, 2014); the framework to understand the needs of logistics at different stages of a crisis and how to meet them (Jahre et al., 2009); the framework defined by stochastic programming in two stages to support transportation planning in disasters (Barbarosoglu & Arda, 2004), and the framework to support purchasing auctions for multiple buyers in humanitarian supply chain management (Ertem et al., 2010).

According to Tomasini & Van Wassenhove (2009), it is important to have systems and processes in place that can facilitate collaboration between the actors to maintain high levels of efficiency when responding to disaster situations. Depending on the maturity of the LH processes, BPM can support humanitarian organizations and beneficiaries, from simple organization to automated control of tasks aligned with strategic objectives.

Considering the complexity of humanitarian operations involved in disaster management, it is essential to use frameworks to support Humanitarian Supply Chain Management

(HSCM), which can be conceptual and/or empirical. Thus, Vaillancourt (2016) proposes a theoretical framework to better understand the incentives and obstacles to the consolidation of materials in humanitarian logistics. As well, Jahre, Jensen, and Listou (2016) state that there is a lack of empirical studies in the area of HSCM frameworks and that it is important to find out the relative contribution of academics, practitioners, and consultants.

Agarwal et al., (2019), meanwhile, point out challenges for future frameworks regarding the importance of coordination among humanitarian actors and measuring the performance of the humanitarian supply chain. Anjomshoae et al., (2017) state that the Balanced Scorecard (BSC) has received considerable interest among practitioners and, unfortunately, its application in humanitarian supply chains presents the lack of causal relationships between performance indicators, actions, and outcomes. Therefore, future research trends focus on unified frameworks to facilitate the discovery of a set of components and elements that support humanitarian logistics operations.

Following these trends, we present here a framework based on an extension of the Reference Task Model (RTF) to support the coordination of humanitarian tasks among actors and a strategic component that concentrates causal relationships aligned with these tasks. It is important to highlight that the design of the strategic plans and the RTF were adapted to the Brazilian reality with information and interviews at the main organization responsible for disaster relief in Santa Catarina, Brazil, called Civil Defense of the State of Santa Catarina (DCESC), together with collaborators, technicians, and academics.

The proposed framework of this work is BPM considered as a management tool that integrates an organization's strategies and objectives with expectations and needs of beneficiaries through end-to-end processes to potentially automated processes (ABPMP, 2013).

From the management point of view, the BPM can be characterized as a continuous life cycle (process) of integrated activities. While several variations of the BPM lifecycle are recognized, most can be summarized by a gradual and interactive set of activities. In the literature there are many "life cycles BPM" and most of them take cyclic forms; thus, it contains a series of actions that will repeat in the next phase (Dumas et al., 2013).

Most of these cycles assume that the organization already has a structure ready to implant the BPM, whether formal or not. In this case, formal denotes if it is included in their orga-

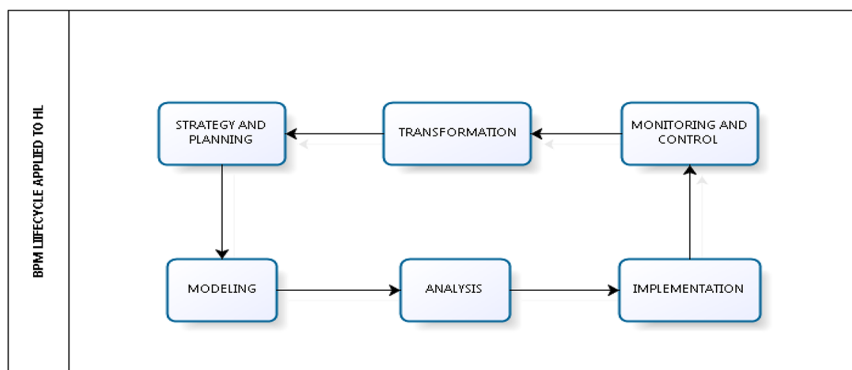


Fig. 2 BPM lifecycle applied to HL

nizational procedures. It should be noted, however, that no “BPM life cycle” corresponds to reality. It is impossible to predict, from a simple theoretical framework, how the BPM will be implemented. This BPM procedure serves as a valuable guide to practice, but the people who implement BPM make all the difference in its application (ABPMP, 2013; Smith, 2007).

To speed up the modelling/design processes, it may be useful to adopt reference models to define the main processes of the organization. In this way, experiences can be leveraged and existing resources in the world can be collected so organizations can find and employ best practices (ABPMP, 2013). In this way, considering the “BPM lifecycle” proposed by ABPMP (2013) and focused on the RTM model proposed by Blecken (2010), Fig. 2 illustrates a BPM procedure applied to the HL processes.

Next, each stage that makes up the proposal of the BPM life cycle applied to the HL is described. This procedure can be applied to a particular process, as well as to integrated management of the entire range of the organization processes, existing or future.

- **Strategy and Planning:** Developing strategic plans for activities in the humanitarian supply chain, of assessment, procurement, warehousing, and transport as the RTM model, from the strategic tool Balanced Scorecard (BSC). It is recommended to consider the prospect: Beneficiaries, Financial Control, Internal Processes, and Learning and Growth, to achieve consensus on objectives, indicators, targets, and actions to be defined for a disaster situation (Schulz & Heigh, 2009; Souza et al., 2015). Listing the HL processes for future disaster situations provides a focus on previous ones.
- **Modeling:** Representation and documentation of how the end-to-end works occur in the tactical and operational plan activities of assessment, procurement, warehousing, and transport of the RTM, to deliver products/services to disaster situation beneficiaries. Blecken (2010) describes in detail these activities and specifies their respective tasks. A simple model might be enough in some cases while a full and detailed model might be necessary for others.
- **Analysis:** Understanding the current processes in the context of desired goals and objectives. This analysis assimilates information from strategic plans, process models, changes in the external environment caused by the nature of the disaster, or other elements of the humanitarian logistics chain.
- **Implementation** may take several forms; one may be a coordinated implementation of process management practices that can be enabled by technology. The array of responsibilities proposed by the RTM model allows connecting activities to people or resources to verify the assignment of duties and responsibilities of the tasks for the parts of the humanitarian supply chain. Implementing but firstly prioritizing processes requires immediate action.
- **Monitoring and Control:** Defining instruments for the measurement and monitoring of processes from indicators defined in the BSC. From this information managers can set resources and timelines to achieve the goals initially set.
- **Transformation** means finding the best way for a process to carry out its work. It might be the usage of new equipment, new practices, new Information Technology (IT) infrastructure, new approaches defined for the HL, or new capabilities.

To our knowledge, there is no literature of a design of BPM lifecycle applied to HL, or supplier choice methods specifically designed or applied to order planning in the humanitarian context. Studies that deal with BPM in HL (see e.g., Blecken 2010; Widera & Hellingrath, 2011) present a general framework of activities. On the order planning or acquisition of products in humanitarian operations, the work developed by Trestrail et al., (2009) described a decision-support tool to improve sea transport and bidding price strategy of food suppliers, and the study by Falasca & Zobel (2011) developed purchasing decision models for humanitarian aid. Both studies focus on specific situations, and a general guideline for the use of quantitative methods in relief operations to assist in the process of acquisition of relief items was not found in the literature.

The frameworks used for case studies in humanitarian logistics proposed in the literature can be classified as generic disaster logistics management, strategic character, and hybrid strategic and generic measures. Specific disaster logistics management identify measures triggered by a concrete disaster with a directed focus on an affected community, as suggested by Tufinki (2006) and Blecken (2010). A generic disaster logistics management, such as that published by Kirac & Milburn (2018), presented a framework for investigating whether it is worthwhile to act on user-generated data before its absolute verification in the context of disaster response logistics planning. The framework described a formal method for quantitatively assessing the value of social data for a variety of disaster response logistics planning problems. It was developed from a case study in the 2010 Haiti earthquake using real social data. Ozguven & Ozbay (2015) proposed a new integrated disaster management framework that can be used for online emergencies. This framework, based on radio frequency identification devices (RFID), is evaluated based on the case study of superstorm Sandy after it hit the coastal region of the U.S. northeast as a post-tropical cyclone. Grass & Fischer (2020) presented a structured framework for case study design; they split the process into different steps and phases. The framework is applied to an illustrative example, where case studies with different numbers and levels of detail of scenarios are designed based on the three-day forecast for hurricane Harvey in 2017. Using empirical data collected for case studies from Australia's "Black Saturday" bushfires in the state of Victoria in February 2009 and from Cyclone Larry in March 2006, Oloruntoba et al., (2018) proposed a framework for examining the preparedness and recovery phases of disaster management activities and processes about predictable disasters within a developed country. Olaogbebikan & Oloruntoba (2019) demonstrated conceptual and theoretical similarities between disaster relief chains (DRCs) and commercial supply chains using the customer relationship management process model (CRM) of the seminal Global Supply Chain Forum framework (GSCF). The authors analyzed a range of empirical data from a case study of the 2006 relief and recovery response to Cyclone Larry disaster using the CRM process model of the GSCF framework as a lens of analysis.

In the case studies researched for generic disasters, several phases were identified, such as regional clustering and risk analysis, pre-assessment, planning and implementation of network structure, and generic contingency supply management. Thus, the definition of strategic objectives, indicators, targets, and actions proposed from an integrated process model to facilitate collaboration among stakeholders and maintain levels of efficiency in response time in disaster situations, is an approach for the activities in the humanitarian supplies chain under the four BSC perspectives proposed in the framework of this article.

Specific disaster logistics management investigate case studies and focus on the uniqueness of the region, test and analyze the application of proposed frameworks for the specific disaster and region. Lu et al., (2019) developed a framework on agility-building strategies used by Humanitarian Organizations (HOs) for the ramp-up. A multiple-case study was conducted on six international HOs that operated in Indonesia. The case study showed that the human resource management, pre-positioning, standardization, and supplier management of the HOs are all related to their resources and environment in the ramp-up process. Given that a single country was the site of study, some findings may not apply to the other HOs or in other regions.

Yilmaz et al., (2019) presented a framework based on Geographic Information System (GIS) that considers widths and structures of different types of infrastructure (roads, tunnels, and bridges). They examine the survivability of infrastructure according to its spatial proximity to a potential disaster and consider legal speed limits in the pre-and post-disaster decision processes. A case study was carried out for an earthquake disaster scenario in Erzurum province of Turkey and various scenarios were generated to analyze the solutions under different conditions.

To incentivize suppliers to reduce delivery lead times, Wang et al., (2019) suggested using a bonus contract as a supplemental agreement to the Fixed Framework Agreement (FFAs). They developed a performance measurement model using Deprivation Level Functions (DLFs) to measure the efficiency and cost-effectiveness of an HL operation. The models were tested in a case study with data collected from Ya'an City, China, which experienced a 7.0 earthquake in 2013. Hasani & Mokhtari (2018) proposed a comprehensive scenario based on the framework. Thus, a proposed mixed-integer linear programming model, one that aims to minimize the total cost of network management, was shown in a real case study of disaster management in Iran. Zhang & Cui (2021) presented a case study with data from Gulf Coast hurricane threats in which they demonstrated the application of proposals in a scenario-based stochastic program. Their study integrates the decisions of prepositioning facility locations, quantities of stocked emergency supplies, service allocations in each scenario in the same modeling framework, and the significance of considering victims' welfare loss in humanitarian relief logistics.

Gonzalez-Feliu et al., (2020) provided a framework to analyze the maturity of humanitarian logistics systems to face crisis situations related to recurrent events and presented the development of three case studies. Two of those studies derive from May 2015: Colombia's mudslide in Salgar, Antioquia and the landslide/river overflow in the Arauca region; the third case study, the Chosica flood, occurred in February-March 2017 in Lima, Peru. The administration of donations, design of a distribution network, and the choice of suppliers are highlighted in that study, as are the main elements that characterize capability in the maturity model for humanitarian logistics. Timperio et al., (2020) proposed an integration of multi-criteria decision-making, network optimization, and discrete event simulation, to address inventory prepositioning for improving efficiency, effectiveness, and agility of relief chains. The authors demonstrated the applicability of the proposed decision support framework in a real-life case study centered in Indonesia, one of the countries with the highest disaster risk exposure on a global scale; their work utilized Logistix software. Khoshsirat et al., (2021) presented a multi-attribute reverse auction framework to facilitate purchase operations between suppliers and a relief organization. The applicability and validity of the proposed auctioning framework were tested through a real case study in Iran. Scholten et

al., (2014) developed an integrated supply chain resilience framework and used an in-depth qualitative case of a collaborative agency. The value of the framework is demonstrated by a retrospective analysis of aid operations in response to Hurricane Katrina. Timperio et al., (2017) provided a decision support framework to identify locations to address network design in the domain of disaster relief supply chains. They used geographic information system technology and fuzzy analytical hierarchy process, and the solution approach was applied to a real-life case about Indonesia. Kinaya et al., (2018) developed and tested a new modeling framework for the shelter site location problem under demand uncertainty. The authors used a maxmin probabilistic programming model that includes two types of probabilistic constraints, and they solved the stochastic shelter site location problem at hand using data from the Kartal district in Istanbul, Turkey.

Whereas the specific-disaster logistics management consists of the operations support process chain, the main focus is the provision of the basic infrastructure needed for the execution of logistical measures. Integrated project planning and control process chain serves as an integration platform during the project and helps to implement the strategic objectives determined in the assessment and pre-assessment phases and in the disaster-specific supply (Tufinki, 2006; Blecken 2010). In the case studies analyzed, various logistics functions were implemented to meet the specific demand of the site affected by the disaster with the use of frameworks. Thus, in this article, tasks are coordinated in a way to assist the relations between logistics functions and the stakeholders involved. We seek to manage the process according to the specific disaster studied.

3 Case Study

3.1 Case Study Description

The state of Santa Catarina has been affected by a wide range of natural disasters, including severe droughts, major floods, and tornadoes. While the region represents only 1.2% of the national territory, Santa Catarina records 12.2% of all natural disasters that occurred in Brazil between 1991 and 2010. That places the state as third in the country's ranking behind only the states of Rio Grande do Sul and Minas Gerais, where these natural phenomena occur even more frequently (Zimmermann, 2011).

The 1983 flood was maintained over a one-month period at levels above 8.5 m, during which four flood waves occurred. The August 1984 flood occurred between five days (August 6th to 10th). According to an analysis developed by Tachini et al., (2009), the damages of August 1984 were reduced by 29.7% as compared to July 1983. The authors also point out that, in addition to the time difference between these events, three facts significantly favored the difference in losses. First, the National Weather Services issued early warning bulletins to the region. Secondly, the Itajaí Valley Hydrometeorological Monitoring System was being tested in August 1984, with three telemetric stations installed. Finally, the Forecasting and Warning System (CEOPS) started its activities with the issuance of the first level forecasts to the municipality of Blumenau, SC.

Blumenau is a municipality in the state of Santa Catarina, southern Brazil, and it is the host city of the Itajaí Valley Region. The third most populous city in the state, the 8th in

southern Brazil, is considered a medium-sized city of Santa Catarina, making it one of its primary industrial, technological, and university centers in the state.

Considering the pivotal role of the Santa Catarina's 1983 event, a case study was developed based on this scenario. The datasets were updated for the official last year of the IBGE (Brazilian Institute of Geography and Statistics) Census, 2010.

Table 1 presents updated data for the top ten most representative cities. It considers the same proportion of homeless people in 1983, so we calculated a newer number of homeless.

Table 1 Year 1983 updated for year 2010

Cities	1983		2010		
	Population	Number of Homeless	Population	Number of Homeless	Homeless (%)
Araran-guá	37,139	1000	62,308	1676	2,69
Blumenau	170,491	50,000	309,214	90,600	29,3
Caçador	42,534	2700	79,088	5014	6,34
Campos Novos	44,144	1080	32,824	801	2,44
Cano-inhas	49,313	5023	52,937	5347	10,1
Chapécó	92,982	1280	198,188	2715	1,37
Correia Pinto	12,625	1050	14,794	1229	8,31
Gaspar	28,012	3981	59,728	8481	14,2
Ireneópolis	9014	2572	10,450	2981	28,53
Itajaí	94,449	40,000	183,388	77,573	42,3
Itapiranga	26,700	1232	15,409	710	4,61
Ituporanga	18,149	1820	22,255	2226	10
Lontras	7390	4000	10,244	5542	54,1
Navegantes	15,747	3070	60,556	11,802	19,49
Porto União	28,254	1700	33,493	2013	6,01
Rio do Oeste	7280	2820	7090	2744	38,7
Rio do Sul	38,616	25,000	61,198	39,620	64,74
Rio Negrinho	22,822	6090	39,846	10,631	26,68
Taio	18,809	5079	17,260	4660	27
Timbó	19,368	1610	36,774	3056	8,31
Três Barras	12,729	3752	18,129	5343	29,47
Trombudo Central	7404	2980	6553	2637	40,24
Xanxerê	31,869	1955	44,128	2705	6,13

The usage of empirical data on HL is relevant because disasters occur in dynamic environment, they are subject to significant changes, and, most of the time, they are unexpected. These data are generally used as a substitute for the actual data and can be used to provide a test environment, which fills the specific requirements.

Table 2 presents the input data relating to the quantities of basic food parcels demands by the city in the requested period. These data were obtained using the projected number of homeless people for the year 2010, as was shown in the previous table and, according to the Civil Defense Department of Santa Catarina, estimates one food parcel for every four homeless people for 15 days. This way, the data on the amount of demand by cities was obtained.

In the considered scenario, the severity of the event and the number of affected cities in the state made it impossible for local suppliers to be available to meet the required demand; therefore, suppliers from other states in Brazil were considered. These providers may be located within a large or a small distance from the affected area. What is sought are those suppliers with product availability closer to the requested time.

Table 3 exemplifies the empirical data used for the computational implementation relating to the supplier identification, represented by the vendor ID, postal code (Zip Code), the date associated with the capacity (per unit) informed by each supplier, and price. It should be noted that the period considered is day and the price (u.m.) per unit of food parcels.

3.2 The use of BPM in the case study

The framework to support the selection of suppliers of humanitarian assistance items in a flood that occurred in Brazil can be observed in Table 4. The strategic plans for the suppli-

Table 2 Product Requested Quantity

Demand (Dm)	Cities	Requested date	Quantity (unit)
1	Araranguá	12-Jul	250
2	Blumenau	20-Jul	2500
3	Blumenau	10-Jul	22,650
...	...	...	...
6	Campos Novos	17-Jul	270
...	...	...	...
29	Xanxerê	13-Jul	489

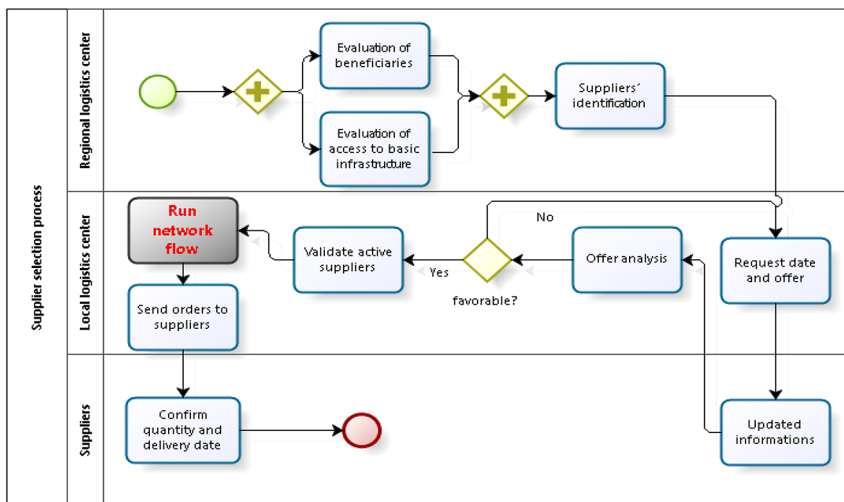
Table 3 Suppliers Empirical Data

Identification Supplier	Name	Zip Code	Date	Capacity (unit)	Price (u.m.)
66	BX	8,943,000	15-Jun-83	200	...
...	...	...	...	...	...
62	BS	300,000,001	21-Jul-83	3200	590
63	BT	30,000,001	22-Jul-83	1100	575
...	AS	89,824,000	18-Jul-83	230	...
...	...	...	...	...	...
48	BE	80,530,000	29-Jul-83	5000	220

Table 4 BSC to suppliers' selection

Perspectives	Strategic Goals	Indicators	Targets	Proposed Actions
Beneficiary	Attending beneficiaries' needs for region in shorter time.	- Disaster average by cluster *	Identifying the type of disaster by region.	Analyzing historical data to form disaster groups by region.
Financial Control	Setting budget by region.	- Product cost - Deprivation Cost**	Displaying products by type of disaster.	Programming a list of items that contain the products used more often in a specific humanitarian operation.
	Identifying the maximum amount and the type of product by disasters.	- Demand met a day	Setting a regional and local supply plan.	Registering local and regional suppliers depending on the type of items per disaster.
Inter Process	Setting the strategic logistics in the supplier selection process for acquisition.	Time index per service a day **	Defining the set of suppliers.	-Setting a deprivation cost, defined as quantitative assessments of human suffering due to lack of access to a good or service -Determining the suppliers' set respecting the time to demand and delivery capacity on the affected site.
Learning and Growth	Manage the good relationship between humanitarian and private organizations.	- Organization size - Segment (sector) - Classification by cluster - Location	Training operators in a disaster situation and the usage of online development tools and relationship systems with suppliers.	Developing a general framework of tasks identifying the specific contributions from the organizations (humanitarian and private) in the acquisition process.

*Lima et al., (2014); **Lima et al., (2015)

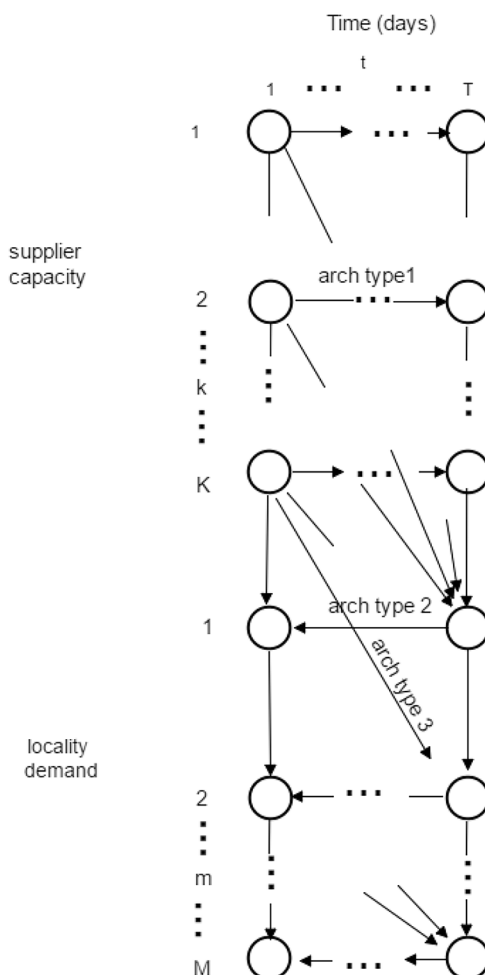
**Fig. 3** Supplier Selection Process

ers' selection are presented under four perspectives: strategic goals, indicators, targets, and proposed actions for each of these perspectives.

In modelling, for the representation and documentation of the involved activities in the selection of suppliers the RTM was used. Figure 3 illustrates the process of selecting suppliers and it considers the existence of some communication between the responsible organizations that serve during disasters (local or regional civil defense) and the suppliers' considered partners. A decision on which suppliers will be chosen to meet the required demand occurs in the activity "Run Network Flow," where the best set of suppliers are defined respecting time, demand, and delivery capacity in the affected area.

In the phase concerning analysis, it was observed that the acquisition from global suppliers is more practiced after the beginning of the disaster, major humanitarian organizations send their workers to the disaster area, and they evaluate people's needs. Depending on the disaster severity level, they decide if the purchases will be performed locally or globally. As

Fig. 4 Graph proposed



the infrastructure is usually destroyed after the disaster, local suppliers may not have easy access.

Due to the intrinsic complexity of the involved factors in HL operations, it is necessary to adopt two considerations: (a) the existence of a partnership between private companies and humanitarian organizations that somehow commit themselves to inform and update their identification data, location, supply capacity, and price; (b) the usage of the system Cost Insurance and Freight (CIF) - transport is the supplier responsibility.

The success of the process illustrated in the previous figure depends on the activity “Run network flow” and this occurs by using the “flow problem” on networks adapted to HL.

The solution will move to choose suppliers who deliver the supplies as soon as possible, even at a higher cost because from the point of view of lives at risk, it is the best option facing a cheaper supplier that will not be able to meet the demand on the requested date. Thus, the “Flow problem” must contain the cost missing for the day not attended (defined here empirically), so that the solution procedure fetches those lower cost suppliers, with product availability at the requested time, or closer to it.

Figure 4 shows the developed graph. In the graph, the node label (supplier capacity, time) represents the amount of available product for immediate delivery at that period. The node label (location demand, time) represents the necessary amount of product at the disaster site in the requested period.

There are three types of arches that represent costs: arch type 1, capacity costs (associated with inventory); arch type 2, deprivation costs, (cost of missing for the day not attended); and arch type 3, flow of the purchase (associated with the product costs).

The simplex method is specialized to solve the flow problem on networks; it is based on the classical structure of Kennington & Helgason (1980), and it was then implemented. The optimized network flow implementation was developed in Java framework with the MySQL database. The tool Anylogic integrated interface was used in the app developed to obtain graphically the sought results. Figure 5 illustrates the technological scheme to support the developed tool.

In the phase concerning monitoring and control, using the network flow problem, the suppliers can be identified who should meet the demand within seven days. Figure 6 shows the tool environment implemented to monitor the analyzed scenario.

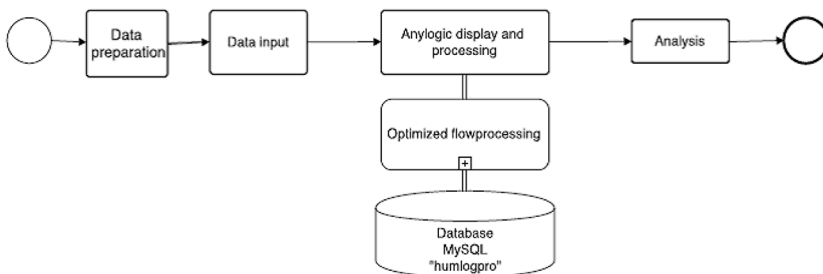


Fig. 5 Technological tool developed scheme

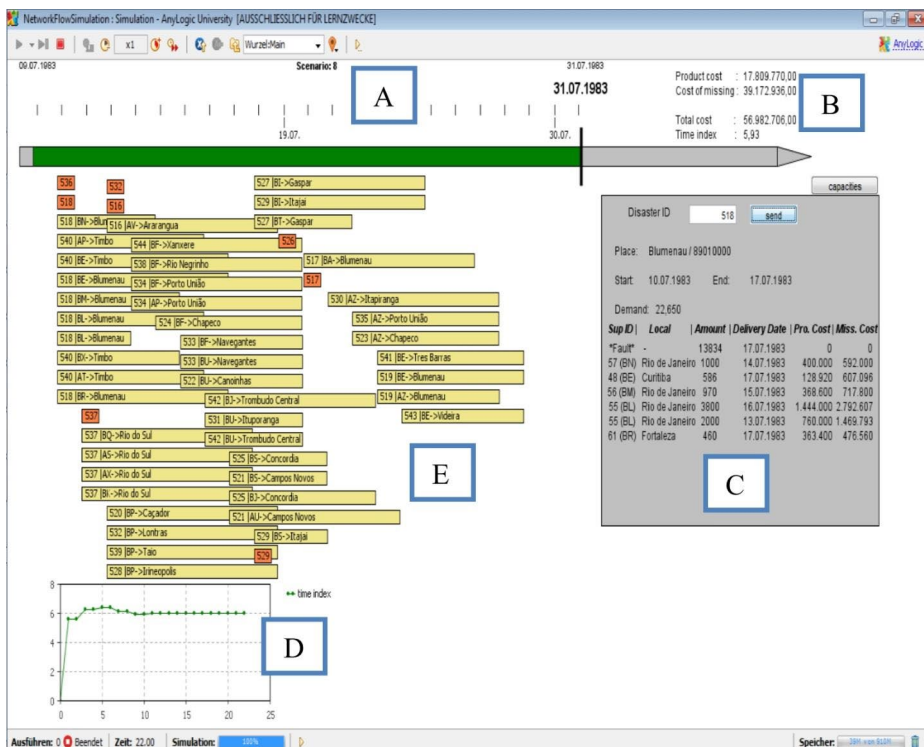


Fig. 6 Monitoring of selected suppliers

The environment illustrated in the previous figure shows some resources to support the monitoring indicators, namely: Time Ruler (A), values of costs (product cost, cost of missing or deprivation costs, and total cost), and performance index through the attending time average (B), picture of specific care by demand location results (C), Graph of daily service time index (D), Supplier service time ruler (demand Id, supplier name, the demand place) (E).

Table 5 presents the results related to 764 units of basic food parcels, requested on 10/07 for the city of Timbó (Santa Catarina/Brazil). These requests were met by partner suppliers with a lower average time of 5 days (20 units were provided by the supplier BX) and the highest average service time was 7 days (614 units were supplied by the supplier of BE).

Generally, suppliers can be located at a large or a small distance from the affected area. The CIF system (Cost, Insurance, and Freight) is used, and the relation between the distance with the transport costs are included in the product price. For example, a supplier located in the northern region of the country is much further than another one located in the southeast, as the affected area was in the south of the country. In this case, considering the transport cost, the final price of the supplier located in the northern region of the country will be higher than the one located in the southeast region. It is worth noting that, the proposed model, the price condition does not make it impossible that the northern region supplier might be chosen. The model searches for the supplier who can meet the demand at the requested time and not at lower cost. Among those suppliers who can meet in requested

Table 5 Requests met by suppliers

Supplier	Location	Supplied quantity	Delivered date	Product cost	Deprivation cost
AP	Chapecó	30	16.07	4800	26,640
BE	Curitiba	614	17.07	135,080	636,104
BX	Águas Frias	20	15.07	4000	14,800
AT	Florianópolis	100	16.07	11,500	88,800

time, the model will choose the one with lower cost. So, if the southeast region supplier has the demand for the requested day as well as the northern region supplier, but the first one offers a lower price, despite the distance, logically will be chosen.

Each city's needs were met according to the availability informed by each partner supplier. Considered were the cost of lack per day (deprivation cost), the program assisted in the nearest to the demand date, and the supplier with sufficient capacity to meet the date requested or the one will availability closest to that date.

Figure 7 shows the average time to meet the demands arising during the considered period.

It can be observed in Figure 7 that 21% of products ordered on July 10th were met three days later, 10% were met in 4 days, 10% required 5 days to be met, 41% of these demands were met six days after, and 17% in 7 days. The average time service for this date was 5 days. On days for July 11, 12, 13, 20, and 21, 100% of requested products have been met only seven days later. On days for July 14 and 22, 100% of the demand was met six days later, and on July 15 and 24, 100% of the ordered products were delivered 5 days later. It is observed in this scenario, in general, that no demand was met at the requested day, and the service time was about 6 days.

Table 6 presents the results related to the time index, product cost, and total cost, considering the variations between capacity and demand.

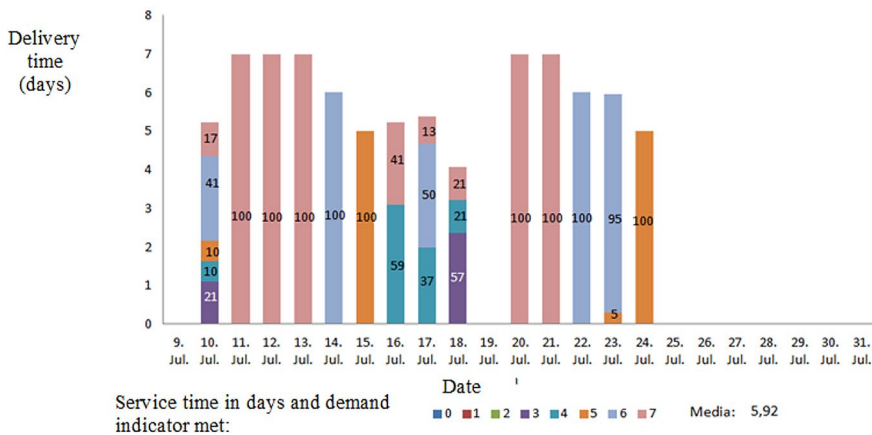
**Fig. 7** Average time of service demands

Table 6 Requests Met by Suppliers

Results	Capacity Supplier (units)	Demand (units)	Cap\dem (units)	Time index (days)	Product Cost (currency unit)	Total Cost (currency unit)
1	541	81.838	0,007	5,76	247.400	687.050
2	25.045	81.838	0,306	5,71	9.992.298	29.698.548
3	51.877	81.839	0,634	5,64	22.850.533	62.816.683
4	75.251	81.839	0,919	5,34	23.810.640	71.230.440
5	99.161	81.839	1,212	4,93	29.502.155	84.218.255
6	118.749	81.840	1,451	4,5	31.061.067	84.887.217
7	174.536	81.841	2,133	2,15	28.738.119	54.291.669
	200.518	81.842	2,45	1,41	30.312.633	469.539.333
9	225.164	81.843	2,751	1,02	38.669.974	52.447.324
10	247.690	81.844	3,026	0,75	23.906.714	33.507.764

In Table 6, it can be observed that the capacity of the supplier is increased randomly. It calculates the variation between the capacity of supply and the quantity of the demand, since the quantity of the demand is fixed and the capacity of supply increases. Thus, the relationship between time index and capacity/demand variation can be evaluated. The higher is the relationship between the capacity and demand, the lower is the time index. Also, the greater the value of the deprivation costs per day, the time index (the number of days to meet the demand) decreases. This characterizes the importance of the deprivation costs per day in the proposed model; that means when this value is well dimensioned, the tendency is to seek a supplier with delivery capacity as close as possible to the requested date, and lower costs should still be considered.

Considering that, when disasters happen, by the characteristics of the problem analyzed, the vendor selection process could be automated from a BPMS tool (Business Process Management System) and a web interface prioritized. All partner suppliers would use instant messaging to update information and to find viable solutions. The supplier registration must be rearranged for each situation, being sorted by increasing order from the practiced prices to the location reached. In the transformation phase, in general, Blecken (2010) confirms that the processes of the supply chain are not yet fully documented and systems based on IT, which could be used to accelerate, automate, and measure these processes, were not properly described and certainly not evaluated.

4 Discussion

According to Behl & Dutta (2019), there are studies referring to Supply Chain Management in Humanitarian Operations, which present HL as a critical theme. They are developed according to the most diverse types of research methodologies, such as theoretical, case studies, mathematical models, research focusing on the properties of the humanitarian supply chain, and resources for the effective management of humanitarian operations.

The objectives served by mathematical models include minimum distribution time, deprivation costs unmet demand, decision-making time, assistance distribution cost, deviation from the expected plan, logistics costs, environmental and operational risk costs,

psychological costs, response time, maximizing reliability and equity, satisfaction, route reliability, demand satisfaction, collective resilience of survivors during emergency logistical operations, and equipment use (see Alem et al., 2016; Anparasan & Lejeune, 2017a, b; Clark & Culkin, 2013; Dewangan et al., 2016; Esmaceli & Barzinpour, 2014; Falasca & Zobel, 2011; Gupta et al., 2016a, b; Habib & Sarkar, 2017; Haghi et al., 2017; Luras et al., 2014; Lima et al., 2015; Nagurney et al., 2016; Ransikarbum & Mason, 2016a, b; Voyer et al., 2015; Zokaee et al., 2016).

This case study contributed to the selection of suppliers based on a framework, considering the best balance in the allocation of supplies, depending on the delivery time, capacities, and deprivation costs, to guarantee the indispensable material resources in the most efficient way.

In this case study, a BPM procedure is adopted to support the supplier selection process of a flood disaster. Each stage that makes up the BPM life cycle proposal in HL is described. Some tools such as the Balanced Scorecard (BSC) are used to obtain consensus on objectives, indicators, goals, and actions to be defined for a disaster situation. To balance supply allocation, the network flow problem was adapted to the quantitative model, containing the variables time, demand, capacity, and the set of suitable suppliers. For the case study, a description of a flood disaster in the State of Santa Catarina, located in southern Brazil, is presented. The main contribution is defined in the result set of an optimized flow and aligned with the strategy of a humanitarian organization.

This article presents a framework for the selection of suppliers or suppliers' groups, using the network flow model, considering the BPM life cycle. The study is limited to the existence of a predetermined group of suppliers, which should be included in the management process of the humanitarian organization, considering that the network flow models traditionally seek lower costs. In the case under study, deprivation costs were included, i.e., among the available suppliers, those who deliver in the shortest time will be chosen. After all, the priority is to meet the demand for disaster as quickly as possible, the model allows balancing according to time and quantity.

The use of the Balanced Scorecard (BSC) for the definition of suppliers (Fig. 2), in the item referring to Strategy and Planning in the BPM life cycle, helped to define the indicators for the four perspectives (Beneficiary, Financial Control, Inter Processes, Learning and Growth), which are included in the network flow model. For example, in the perspective referring to Internal Processes, the service time index per day was represented in the Service time index graph per day, (item D in Fig. 6). The Financial Control perspective in the figure indicates product cost, deprivation cost, and demand served per day with the cost values and performance index, through the average service time (item B of Fig. 6).

In this way, the strategic objectives, goals, and actions suggested in the BSC are directly linked to the proposed model, in which to determine the best set of suppliers, it is necessary to indicate the type of disasters by region, to identify the products by type of disasters, to define a regional and local supply plan and a set of suppliers, and finally, to train employees in a disaster operation to use online development tools and supplier relationship systems.

5 Conclusions and Recommendations

A characteristic of the proposed framework is that it focuses on natural disasters in the southern region of Brazil (floods, landslides, and tornadoes) and concentrates the experiences and procedures of the main organization in the state of Santa Catarina, DCESC. The framework was designed empirically with the participation of technicians and academics. The central component of the proposed framework is the RTM model that can be adapted and/or extended to any humanitarian organization. Here, for example, the rigid procurement tasks of the humanitarian organization were made flexible for emergency situations, respecting the norms of the organization. Consequently, it was necessary to monitor these tasks and to control them, making it necessary to define a strategic component that aligns these tasks with indicators. Also in the supplier selection task, the pre-positioning strategy was incorporated, which consists of defining the deprivation cost, attributing a value here to the lack of humanitarian items per day, and supporting the tendency is to optimally search for that supplier with the capacity to deliver as close as possible to the requested date. Considering the characteristics of the proposed framework, its implementation can be realized in a simple way from process-based technologies and from those that are strategy-focused.

According to an analysis of the literature presented by Jabbour et al., (2019), research gaps, fields of interest, and trends for the HLSCM study field are in evidence, including such items as the establishment of public-private partnerships for the involvement of suppliers who should be organized to support the preparation and prevention of natural disaster situations (and of human-made disasters as well), and the finding of big data to make quantitative studies viable in this field of research. In the case study, the public and private sectors are fundamental, mainly regarding the updating of the records. Further, communication of their availability and other factors are linked with big data use.

This article presents a framework defined by a BPM procedure and employs the RTM model as a reference. This procedure was applied to HL to support the supplier selection process from flow-optimized networks. It was defined as a simplified BPM lifecycle and recommended for humanitarian organizations in the structuration phase. The proposal was applied for a flood scenario that occurred in the State of Santa Catarina, Brazil.

The approach-related steps in the BPM lifecycle showed the strategic plans for the activities in the humanitarian supplies chain under four BSC perspectives. The definition of strategic objectives, indicators, targets, and proposed actions in each of the perspectives demonstrated that having systems and processes can facilitate collaboration among stakeholders to maintain the high-efficiency levels at the response time in disaster situations.

The application of BPM procedure was considered modelling from Process Modelling Notation Standard of acquisition activities at the operational level. It was observed that the activities can help the process, and it can be verified that the transparency of these activities can support the harmonization of relations between the humanitarian organization sectors and alignment with its strategies.

In the operational phase, a flow in optimized networks was analyzed and adapted to humanitarian operations. This flow problem can help to identify suppliers that meet the demand in the shortest time possible, considering a cost of lack per day. This trial was presented several scenarios to evaluate the average time of demand service over the planning horizon. The results presented here are consistent and this tool is useful for humanitarian organizations to prepare and respond to disasters and for better integration between the steps

defined in the BPM procedure. We recommend the usage of a Business Process Management System – BPMS.

For future research it is suggested to model and document the HL processes on humanitarian organizations aiming to solve scenarios of acquisition and transportation to emergencies, thus adding new variables. The improvement of coordination processes of disaster may serve as a basis for government agencies and institutions involved in the coordination of logistic processes in emergency situations.

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